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SUPERSEDING
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MILITARY SPECIFICATION

ANEMOMETER ML-433( )/PM AND CASE, ANEMOMETER CY-1397( )/PM

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers one type of meteorological equipment designated Anemometer ML-433( )/PM and Case, Anemometer CY-1397( )/PM (see 6.4).

2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications, standards, and handbooks. Unless otherwise specified, the following specifications, standards, and handbooks of the issue listed in that issue of the Department of Defense Index of Specifications and Standards (DoDISS) specified in the solicitation form a part of this specification to the extent specified herein.

SPECIFICATIONS

FEDERAL

L-P-378	- Plastic Sheet and Strip, Thin Gauge, Polyolefin.
NN-P-71	- Pallet, Material Handling, Wood, Stringer Construction, 2 Way and 4 Way.

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Commander, US Army Electronics Research and Development Command, Combat Surveillance and Target Acquisition Laboratory, ATTN: DELCS-PE, Fort Monmouth, New Jersey 07703, by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

FSC 6660

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QQ-S-781	- Strapping, Steel, and Seals.
PPP-B-585	- Box, Wood, Wirebound.
PPP-B-601	- Boxes, Wood, Cleated Plywood.
PPP-B-621	- Box, Wood, Nailed and Lock Corner.
PPP-B-636	- Box, Shipping, Fiberboard.
PPP-B-640	- Box, Fiberboard, Corrugated, Triple wall.
PPP-F-320	- Fiberboard, Corrugated and Solid, Sheet Stock (Container Grade), and Cut Shapes.
PPP-S-760	- Strapping, Nonmetalic, (and Connectors).
PPP-T-97	- Tape, Pressure-Sensitive Adhesive, Filament Reinforced.

MILITARY

MIL-P-116	- Preservation-Packaging, Methods Of.
MIL-P-11268	- Parts, Materials, And Processes Used In Electronic Equipment.
MIL-M-13231	- Marking Of Electronic Items.
MIL-F-14072	- Finishes For Ground Signal Equipment.
MIL-P-19644	- Plastic Molding Material (Polystyrene Foam, Expanded Bead).
MIL-P-26514	- Polyurethane Foam, Rigid or Flexible, For Packaging.

STANDARDS

FEDERAL

FED-STD-101	- Preservation, Packaging and Packing Materials, Test Procedures.

MILITARY

MIL-STD-105	- Sampling Procedures And Tables For Inspection By Attributes.
MIL-STD-129	- Marking For Shipment and Storage.
MIL-STD-147	- Palletized Unit Loads.
MIL-STD-169	- Extreme Temperature Cycle.
MIL-STD-170	- Moisture Resistance Test Cycle For Ground Signal Equipment.
MIL-STD-202	- Test Methods for Electronic and Electrical Component Parts.
MIL-STD-252	- Wired Equipment, Classification of Visual and Mechanical Defects.
MIL-STD-454	- Standard General Requirements For Electronic Equipment.
MIL-STD-726	- Packaging Requirements Code.
MIL-STD-810	- Environmental Test Methods.

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2.1.2 Other government documents, drawings, and publications. The following other government documents, drawings, and publications form a part of this specification to the extent specified herein.

DRAWINGS

ELECTRONICS COMMAND

SC-DL-87824	- Case, Anemometer CY-1397( )/PM
SC-DL-106434	- Anemometer ML-433( )/PM

(Copies of specifications, standards, handbooks, drawings and publications required by manufacturers in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting officer.)

3. REQUIREMENTS

3.1 First article. When specified, the contractor shall furnish sample units for first article inspection and approval (see 4.3 and 6.2d).

3.2 Parts, materials, and processes. In addition to the requirements of this specification, the requirements of MIL-P-11268, including the selection requirements therein, shall apply.

3.3 Recycled, virgin and reclaimed materials. Without jeopardizing the intended use of equipment furnished to this specification, selection of materials will be governed by the following:

a. There is no exclusion to the use of recovered materials.
b. There is no requirement that equipment be manufactured from virgin materials.
c. Reclaimed materials are required to the maximum extent possible.

3.4 Construction. Anemometer ML-433( )/PM and Case, Anemometer CY-1397( )/PM shall be constructed in accordance with SC-DL-106434 and SC-DL-87824 respectively.

3.5 Finish. The equipment shall be finished in accordance with MIL-F-14072 and the equipment drawings.

3.6 Marking. Marking shall conform to MIL-M-13231 and the equipment drawings.

3.6.1 Visibility. Wherever practicable, parts shall be so mounted that their identification markings will be readily visible with minimum disassembly of the equipment.

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3.6.2 Serial numbers. Each ML-433( )/PM Anemometer shall be serial numbered.

3.6.3 Facsimiles. Facsimile copies of the conversion chart, velometer scale plate, and compass card, as specified in the contract, shall be submitted to the contracting officer for approval (see 6.2g).

3.6.7 Fungus. Equipment and assemblies shall show no evidence of visible fungus or corrosion when tested in accordance with 4.10. (Corrosion is defined as any visible degradation that can be attributed to flaky, pitted, blistered, or otherwise loosened finish or metal surface.)

3.8 Interchangeability. Like units, subassemblies, and replaceable parts shall conform to requirement 7 of MIL-STD-454 (see 4.11).

3.9 Service conditions. The equipment shall meet the following service conditions.

3.9.1 Equipment nonoperating. The equipment shall comply with the operational requirements of 3.9.2 after subjection to the following nonoperating conditions:

a. Temperature. Continuous exposure for 72 hours at 160°F (71°C) and 72 hours at -80°F (-62.2°C), at any orientation.

b. Relative humidity. Relative humidity up to 97 percent, for an indefinite period of time; and exposure at 100 percent relative humidity, with condensation, for 4 hours. Following subjection to such conditions and a period of 16 hours at normal room temperature and relative humidity, the equipment shall meet the requirements of 3.9.2 (see 4.8.4).

c. Vibration, bounce and shock (see 3.9.3).

3.9.2 Equipment operating. The equipment shall meet the performance requirements of this specification while subjected to any of the following conditions:

a. Temperature. Ambient temperature in the range of 150°F (65.6°C) to -40°F (-40°C); the 150°F (65.6°C) temperature includes effect of sun-load. Exposure at the high temperature extreme not to exceed 4 hours and at the low temperature extreme not to exceed 72 hours (see 4.8.5).

b. Relative humidity. Relative humidity up to 95 percent; and exposure at 100 percent relative humidity, with condensation, for 4 hours (see 4.8.4).

c. Elevation. Elevation up to 10,000 feet (3048m) above sea level.

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3.9.3 Vibration, bounce, and shock.

3.9.3.1 Vibration. Except for internal resonance of parts and subassemblies specified herein, the equipment shall have no mechanical resonance below 55 cycles per second.

3.9.3.2 Bounce and shock. The equipment shall meet the tests as specified in table I.

TABLE I. Bounce and shock tests requirements.

Test	Test method paragraph	Requirements after test
Bounce	4.8.2	Full specification performance. No physical damage except surface abrasion.
Shock: drop	4.8.3	Equipment shall be operable. No physical damage, except minor surface abrasion to case.

3.10 Anemometer ML-433( )/PM.

3.10.1 Velometer. The deflecting vane type velometer shall provide for measurement of wind speed in two ranges: one range for wind velocities of 0 to 8 knots with an accuracy of ±1.0 knot, and the other for wind velocities of 0 to 40 knots with an accuracy of ±2 knots.

3.10.1.1 Wind speed sensitivity. The velometer shall be responsive to wind speeds of 1.5 knots when operated in the 0-8 knots range, and to wind speeds of 7.5 knots when operated in the 0-40 range.

3.10.2 Wind direction indication. The wind direction vane and compass shall provide for measurement of wind direction to an accuracy of ±11.25 degrees.

3.10.2.1 Wind direction sensitivity. The wind direction indicator shall be responsive to winds of 1.5 knots or less.

3.10.2.2 Compass operation. The compass shall operate freely, without sticking, throughout the operating service condition temperature range specified in 3.9.2a.

3.10.2.3 Compass card. The compass card shall be mounted with the scale rotated 90 degrees from the actual compass heading so that the correct wind direction will be read by the operator when the anemometer is held in the hand with the wind direction vane facing into the wind.

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3.10.3 Vane cover. A cover shall be provided for the wind direction vane. This cover shall protect the vane from damage when in place, and shall be secured to the anemometer case with a chain to prevent loss.

3.10.4 Handle. A detachable handle shall be provided for holding the anemometer in operation.

3.10.5 Wind speed correction factor chart. A chart shall be provided for computation of wind speed error readings. The chart shall correlate values of temperature, barometric pressure and wind speed correction factors. The chart shall cover a temperature range of -40°F (-40°C) to 160°F (71°C), and a barometric pressure range of 1100 to 600 millibars. Simplified instructions and an example of true wind speed computation shall be printed on the chart.

3.11 Case, Anemometer CY-1397( )/PM. The case shall protect the anemometer from damage due to shocks or vibrations normally encountered in transit, and shall meet the tests of 4.8.2 and 4.8.3.

3.12 Preconditioning. The equipments shall be capable of meeting the inspection of section 4, without subsequent processing, after subjection to the bounce preconditioning, as specified in 4.6.

3.13 Technical literature, tools, and running spare parts. Technical literature, tools, and running spare parts shall be furnished if specified in the contract (see 6.2i).

3.14 Workmanship. The equipment shall be manufactured and assembled in accordance with requirement 9 of MIL-STD-454 and the applicable portions of the following paragraphs:

a. In MIL-P-11268:

Cleaning:
- Parts.
- Units.

Soldering and brazing.

Welding:
- Cleaning prior to welding process.
- Cleaning after welding.

Tropicalization of materials.

b. In this specification:

Construction (see 3.4).
Finish (see 3.5).
Marking (see 3.6).

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4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the government. The government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.2 Classification of inspection. The inspection requirements specified herein are classified as follows:

a. First article inspection (does not include packaging) (see 4.3).

b. Inspection covered by subsidiary documents (see 4.4).

c. Quality conformance inspection.

(1) Quality conformance inspection of equipment before packaging (see 4.5).

(2) Quality conformance inspection of packaging (see 4.12).

4.3 First article. This inspection will be performed by the contractor unless otherwise specified in the contract. It shall consist of the inspection specified in the subsidiary documents covering the items listed in 4.4, and the inspection for group A, group B and group C testing.

4.4 Inspection covered by subsidiary documents. The following shall be inspected under the applicable subsidiary documents as part of the inspection required by this specification, and the inspection requirements specified in the contract:

Item	Requirement paragraph
Finish	3.5
Marking	3.6

4.5 Quality conformance inspection of equipment before packaging. The contractor shall perform the inspection specified in 4.4 and 4.5.1 through 4.5.4. This does not relieve the contractor of his responsibility for performing any additional inspection which is necessary to control the quality of the product and to assure compliance with all specification requirements.

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4.5.1 Group A inspection. This inspection including sampling, shall conform to table II and the procedures of MIL-STD-105 using the general inspection levels. Group A inspection shall be performed in any order which is satisfactory to the government.

TABLE II. Group A inspection.

Inspection	Requirement paragraph	Test method paragraph	AQL(% defective)	
			Major	Minor
Visual and mechanical				
Anemometer ML-433( )/PM	3.14	4.7	1.0	4.0
Case, Anemometer CY-1397( )/PM	3.11, 3.14	4.7	1.0	4.0
Accuracy of wind speed	3.10.1	4.9.2		
Sensitivity to wind speed	3.10.1.1	4.9.3	1.0	
Sensitivity to wind direction	3.10.2.1	4.9.4	for	
Accuracy of wind direction	3.10.2	4.9.5	the	
Compass operation	3.10.2.2	4.9.5.1	group	

4.5.2 Group B inspection. This inspection, including sampling, shall conform to table III and to the procedures for small sample inspection of MIL-STD-105 using the special inspection levels. Group B inspection shall normally be performed on inspection lots that have passed group A inspection and on samples selected from units that have been subjected to and met the group A inspection. The AQL shall be 4 percent defective and the inspection level shall be S-4 for normal, tightened and reduced inspection.

4.5.2.1 Order of inspection within group B. Group B inspection shall be performed in any order which is satisfactory to the government.

TABLE III. Group B inspection.

Inspection	Requirement paragraph	Test method paragraph
Bounce	3.9.3.2	4.8.2
Shock, drop	3.9.3.2	4.8.3
Interchangeability	3.8	4.11

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4.5.3 Group C inspection. This inspection shall consist of the tests specified in table IV and shall be performed on sample units that have been subjected to and met groups A and B inspection. Sample units shall be selected in accordance with 4.5.3.1.

TABLE IV. Group C inspection.

Inspection	Requirement paragraph	Test method paragraph
Temperature	3.9.1a, 3.9.2a	4.8.5
Moisture resistance	3.9.1b, 3.9.2b	4.8.4
Vibration	3.9.3.1	4.8.1
Fungus	3.7	4.10

4.5.3.1 Sampling for inspection. One sample unit shall be selected at random for all group C inspection in table IV every six months or every 250 units, whichever occurs first. The first sample selected shall be at the start of the contract from the first quality conformance inspection production lot.

4.5.3.2 Group C failures. No failures shall be allowed in group C inspection. If a sample unit fails to pass group C inspection, the supplier shall take corrective action on the material and/or process to correct all units which are considered subject to the same failure. Acceptance of the product shall be discontinued until corrective action, acceptable to the government, has been taken. After corrective action, group C inspection shall be repeated on an additional sample, and final acceptance shall be withheld until the group C reinspection has shown that the corrective action was successful. At the option of the government, group A and B inspections may precede the group C reinspection on corrected units and acceptance deferred until all tests and inspections are successfully passed.

4.5.4 Reinspection of conforming group C sample units. Unless otherwise specified, sample units which have been subjected to and passed group C inspection may be accepted on the contract, provided all damage is repaired and the sample units are resubjected to and pass group A inspection.

4.6 Bounce preconditioning. The equipment with shock mounts (if any) removed or blocked, shall be placed in its normal operating position on the table of the package tester, as made by the L.A.B. Corporation, Skaneateles, New York, or equal. The package tester shafts in phase shall have a speed such that it is just possible to insert a strip of material, .03125 inch (0.8mm) thick, under one corner or edge of the unit to a distance of 3 inches (76.2mm) as the unit bounces. The unit shall be subjected to this preconditioning for 1 minute. After bounce preconditioning, the unit shall not be repaired, alined, cleaned, or otherwise changed prior to subjection to procurement inspection.

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4.7 Visual and mechanical inspection. The equipment shall be examined for the defects listed in MIL-STD-252 and table V.

TABLE V. Classification of visual and mechanical defects.

Classification	Defect
Major	Wind direction vane not properly balanced on pivot. Wind speed orifice adjustment binding or inoperative. Wind speed vane and pointer assembly not operating freely or excessive play or binding in pivots. Compass not properly balanced on pivot (sticking). Handle cannot be fastened securely. Fasteners inoperative by hand. Cover will not close.
Minor	Nuts or screws loose but not causing inoperation. Soldering, brazing or welding adequately strong, but of poor workmanship quality. Nails or screws protruding, possibly causing injury. Handle can be properly secured but threads too tight or too loose. Vane cover chain not properly secured so that there is a possibility of it becoming detached. Felt padding loose. Nails or screws protruding, no danger of injury. Fasteners difficult to operate by hand.

4.8 Service condition tests.

4.8.1 Vibration test. The equipment shall be subjected to the test of method 204C of MIL-STD-202E (see 3.9.3.1).

4.8.2 Bounce test. The equipment shall be subjected to the test of method 514.2, part 2, procedure XI of MIL-STD-810 (see 3.9.3.2).

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4.8.3 Shock, transit drop test. The equipment, in its carrying case (see 6.5.1), shall be subjected to the test method 516.2, procedure II of MIL-STD-810 (see 3.9.3.2).

4.8.4 Moisture-resistance test.

4.8.4.1 Test procedure. The components will be tested as follows:

a. Measure performance to establish standards of performance before placing components in humidity chamber (see 4.8.4.2).

b. Dry at 130° ±5°F (54.4° ±2.8°C) for 24 hours.

c. Condition at 77° ±5°F (25° ±2.8°C) and 40 to 50 percent relative humidity for 24 hours.

d. Components with sealed cases shall not be opened.

e. Subject the components to continuous cycling for two 48-hour cycles. Detail requirements for relative humidity, temperature, and time for each cycle shall be as shown in MIL-STD-170. Repeat performance measurements taken in "a" after the two cycles have been completed and not during the periods indicated by the standard.

4.8.4.2 Performance. Measurements made before and after test, and specified performance after cycling, shall be as follows:

Measurement	Test method paragraph	Requirements after cycling
Accuracy of windspeed	4.9.2	Full specification performance
Sensitivity of wind direction	4.9.4	
Compass operation	4.9.5.1	

4.8.4.3 Failure. If any component fails to meet the performance specified in 4.8.4.2 after cycling, it shall be realigned or readjusted once. If the component then fails to meet 4.8.4.2, it does not pass the test.

4.8.5 Temperature test. The equipment shall be subjected to the temperature cycle shown in MIL-STD-169. At steps 1, 3, 5, 8, and 10 of the cycle, the following measurements shall be taken:

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Measurement	Test method paragraph	Requirements
Accuracy of windspeed 1/	4.9.2	Full specification performance
Sensitivity to wind direction	4.9.4	
Compass operation	4.9.5.1	

1/ For the purpose of the moisture-resistance and temperature tests, the wind speed test may be modified to require tests only at the low range (0 to 8 knots) settings.

4.9 Operational tests.

4.9.1 Wind measurements. The tests of 4.9.2 through 4.9.4 require the use of a wind tunnel capable of producing a controlled flow of air with velocities up to 40 knots. Also required is an air speed measuring device having a ±0.25 knot accuracy and capable of measuring the true air speed over the required range.

4.9.2 Accuracy of wind speed. The anemometer shall be placed in the wind tunnel in its normal operating position and wind speed readings taken (see 3.10.1) at true wind tunnel air speeds as follows:

a. The anemometer shall be adjusted for the 0 to 8 knots range and readings made at 2, 4, 6 and 8 knots.

b. The anemometer shall be adjusted for the 0 to 40 knots range and readings made at 10, 20, 30 and 40 knots.

4.9.3 Sensitivity to wind speed. With true wind tunnel air speeds set 1.5 and 7.5 knots, readings shall be taken on the 0 to 8 knots range and 0 to 40 knots range respectively to determine compliance with 3.10.1.1.

4.9.4 Sensitivity to wind direction. With the wind tunnel air speed reduced to 1.5 knots, the anemometer case shall be rotated approximately 10 degrees to the right and left. The wind direction vane should remain aligned in the direction of the air stream in order to meet the requirements of 3.10.2.1.

4.9.5 Accuracy of wind direction. The anemometer shall be placed in a normal operating position on a horizontal table which can be rotated and which is calibrated in circular degrees. The anemometer and tables shall be oriented with the earth's magnetic field or an artificial magnetic field.

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The table shall be rotated so that the compass indicates the compass points with respect to the magnetic field. The scale of the table shall be read at each point and compared to the reading of the compass which should be accurate within the requirements of 3.10.2.

4.9.5.1 Compass operation. The freedom of compass rotation shall be noted throughout 360° of rotation during the test of 4.9.5, and during the temperature cycling test 4.8.5 to check compliance with 3.10.2.2.

4.10 Fungus test. The equipment and assemblies shall be subjected to test method 508.2 of MIL-STD-810. The equipment shall be subjected to the test in the open condition (see 3.7).

4.11 Inspection for interchangeability. The dimensions listed below shall be gaged or measured to determine compliance with the physical interchangeability requirements of 3.8. When a listed dimension is not within specified or design limits, it shall be considered a major defect.

a. Dimensions of vane cover and mating bracket which affect mating of the cover.

b. Size and form of threads on handle and mating thread on anemometer.

c. Internal dimensions of Case CY-1397( )/PM which affect stowing of anemometer.

4.12 Quality conformance inspection for packaging. Quality conformance inspection for packaging shall be in accordance with the group A, group B, and when required (see 6.2b), the group C (rough handling) requirements of MIL-P-116. Inspection lots shall be in accordance with MIL-P-116.

5. PACKAGING

5.1 Preservation. Preservation shall be level A or B as specified (see 6.2c).

5.1.1 Level A.

5.1.1.1 Cleaning. Each Anemometer ML-433( )/PM shall be cleaned in accordance with process C-1 of MIL-P-116.

5.1.1.2 Drying. Each Anemometer ML-433( )/PM shall be dried in accordance with the applicable procedure of MIL-P-116.

5.1.1.3 Preservative application. No preservative required.

5.1.1.4 Unit packing. Unit packing shall be in accordance with the methods or submethods prescribed in MIL-P-116 as specified herein insuring compliance with the applicable requirements of that specification.

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5.1.1.4.1 Technical literature. Each technical literature shall be unit packed submethod IC-1.

5.1.1.4.2 Anemometer ML-433( )/PM. Unit pack each item in accordance with MIL-STD-726, coded as follows:

3G-1-1-ØØ-00-NA-X-ED-Ø-00-A.

5.1.1.4.3 Case, Anemometer CY-1397( )/PM. Unit pack each case in accordance with MIL-STD-726, coded as follows:

1Ø-1-1-ØØ-00-NS-X-ED-Ø-00-A

5.1.1.4.4 Anemometer ML-433( )/PM in Case, Anemometer CY-1397( )/PM. Unit pack each method III as follows: Cushion all surfaces with cells or pads or both fabricated from fiberboard conforming to PPP-F-320, class weather-resistant, designed to protect the unit or packaging components from physical damage during handling, transit and storage. As an alternate, cushioning material may be fabricated of plastic molding material conforming to MIL-P-19644 or polyurethane foam conforming to MIL-P-26514 or any other cushioning material complying with federal or military specifications which will provide the protection specified. Place into a close-fitting fiberboard box conforming to PPP-B-636, W5c. Place the technical literature, unit packed as specified in 5.1.1.4.1, on top of the contents, directly under the lid of the box. Closure shall be in accordance with the appendix of the box specification.

5.1.2 Level B.

5.1.2.1 Anemometer ML-433( )/PM. Unit pack each item in accordance with MIL-STD-726, coded as follows:

2E-1-1-ØØ-00-NA-X-ED-Ø-00-B

5.1.2.2 Case, Anemometer CY-1397( )/PM. Unit pack each case in accordance with MIL-STD-726, coded as follows:

1Ø-1-1-ØØ-00-NS-X-ED-Ø-00-B

5.1.2.3 Anemometer ML-433( )/PM in Case, Anemometer CY-1397( )/PM. Cleaning, drying, preservative application, and unit packing shall be as specified in 5.1.1.

5.2 Packing. Packing shall be level A or B, as specified (see 6.2c). Shipping containers for all levels shall be capable of stacking and supporting superimposed loads during shipment and storage without damaging the container(s) or its contents when tested in accordance with methods 5016 and 5017 of FED-STD-101.

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5.2.1 Level A. A multiple quantity of Anemometer ML-433( )/PM in Case, CY-1397( )/PM, or Anemometer ML-433( )/PM or Case CY-1397( )/PM, unit packed as specified in 5.1 and bearing the same stock number, shall be packed within a close-fitting box conforming to PPP-B-601, overseas type; PPP-B-621, Style 4, class 2; or PPP-B-585, Style 2 or 3, class 3. When the gross weight exceeds 200 pounds (90.7kg), or the container length and width is 48 x 24 inches (121.9 x 61 cm) or more and the weight exceeds 100 pounds (45.4kg), 3 x 4 inch (76.2 x 101.6mm) skids, laid flat, shall be applied in accordance with the requirements of the container specification, or if not specified in the specification, in a manner which will adequately support the item and facilitate the use of material handling equipment. Closure and strapping shall be in accordance with the applicable container specification or appendix thereto except that metal strapping shall conform to QQ-S-781, type I, finish A. When the quantity of a stock-numbered item being shipped to a single destination comprises a shipping container volume of less than one cubic foot (.028m^3), packing shall be as specified for level B.

5.2.1.1 Unitization. Palletization shall be required when: containers specified in 5.2.1 do not require skids; quantities per destination exceed either a total of 250 pounds (113.4kg) (excluding the pallet) or a volume of 20 cubic feet (0.57m^3); and the container size permits use of one of the pallet patterns of MIL-STD-147. A quantity of containers, packed as specified in 5.2.1 except that container strapping may be omitted, shall be placed on a pallet, load type I, conforming to MIL-STD-147. The pallet shall conform to NN-P-71, type IV, group I or II woods. The load shall be "bonded" to the pallet by strapping conforming to QQ-S-781, type I, finish A, or shrink film conforming to L-P-378, type IV. Stretch wrap in accordance with MIL-STD-147 is authorized for shipments within the continental United States and for containerized shipments.

5.2.2 Level B. A multiple quantity of Anemometer ML-433( )/PM in Case, CY-1397( )/PM, or Anemometer ML-433( )/PM or Case CY-1397( )/PM, unit packed as specified in 5.1 and bearing the same stock number, shall be packed within a close-fitting box conforming to PPP-B-640, class 2, style E or PPP-B-636, type CF, class weather-resistant. The gross weight of boxes conforming to PPP-B-640 shall not exceed 250 pounds (113.4kg). Closure shall be in accordance with the appendix of the applicable box specification. Reinforcing shall be by pressure-sensitive filament tape banding or nonmetallic strapping conforming to PPP-T-97 and PPP-S-760, respectively; selection of the material and application shall be in accordance with the appendix of the applicable box specification. When the gross weight exceeds 200 pounds (90.7kg), or the container length and width is 48 x 24 inches (121.9 x 61cm) or more and the weight exceeds 100 pounds (45.4kg), containers will be pallet-mounted on pallets conforming to NN-P-71, group I or II woods. The load shall be "bonded" to the pallet by strapping conforming to QQ-S-781, type I, finish A or shrink film conforming to L-P-378, type IV. When a single unit is shipped to a single destination, no packing is required; the unit container shall serve as the shipping container. Reinforcing shall be as specified above.

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5.2.2.1 Unitization. Palletization shall be required when quantities per destination exceed either a total of 250 pounds (113.4kg) (excluding the pallet) or a volume of 20 cubic feet ($0.57m^3$); and the container size permits use of one of the pallet patterns of MIL-STD-147. A quantity of Anemometer ML-433( )/PM in Case, CY-1397( )/PM, or Anemometer ML-433( )/PM or Case CY-1397( )/PM, unit packed as specified in 5.1 and bearing the same stock number, shall be placed on a pallet, load type I, conforming to MIL-STD-147. The pallet shall conform to NN-P-71, type IV, group I or II woods. The load shall be "bonded" to the pallet by strapping conforming to QQ-S-781, type I, finish A, or shrink film conforming to L-P-378, type IV. Stretch wrap in accordance with MIL-STD-147 is authorized for shipments within the continental United States and for containerized shipments.

5.3 Marking. In addition to any special marking required by the contract or order, interior packs and exterior shipping containers shall be marked in accordance with MIl-STD-129.

6. NOTES

6.1 Intended use. This equipment is a deflecting vane type anemometer with an integral wind vane and magnetic compass for indicating wind speed and direction for use with Meteorological Stations, Manual AN/PMQ-1( ) and AN/PMQ-4( ).

6.2 Ordering data. Procurement documents should specify the following:

a. Title, number, and date of this specification and any amendment thereto.

b. When rough handling test is required (see 4.12).

c. Level A or level B preservation and packing (see section 5).

c. Number of first articles to be submitted for approval (see 3.1).

e. Marking and shipping of samples.

f. Place of final inspection.

g. Number of facsimiles required.

h. The contractor should apply for nomenclature in accordance with the applicable clause in the contract.

i. If technical literature, tools, or spare parts are required.

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6.3 Environmental. Environmental pollution prevention measures are contained in the packaging material specifications referenced herein. Refer to material specifications or preparing activity for recommended disposability methods.

6.4 Nomenclature. The parentheses in the nomenclatures will be deleted or replaced by a letter identifying the particular design, for example: ML-433W/PM. The contractor should apply for nomenclature in accordance with the applicable clause in the contract.

6.5 Transit cases. If the anemometer is procured as part of AN/PMQ-4, Case, Anemometer CY-1397( )/PM will not be procured. If the anemometer is procured as a separate item to be used independently, Case, CY-1397( )/PM should be procured for transit and storage of the item.

6.5.1 Transit case for test purposes. The transit case referred to in 4.8.2 and 4.8.3 should be the one used in the particular procurement. For example, CY-1397( )/PM if procured alone, CY-1010( )/PMQ-4 if procured as part of AN/PMQ-4( ), or CY-952/PMQ-1 if procured as part of AN/PMQ-1( ).

6.6 Changes from previous issue. Asterisks are not used in this revision to identify changes with respect to the previous issue due to the extensiveness of the changes.

Custodians:
Army - ER
Navy - AS

Review activity:
DSC - GS

Preparing activity:
Army - ER

(Project 6660-0060)

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(TO DETACH THIS FORM, CUT ALONG THIS LINE.)

STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

(See Instructions – Reverse Side)

1. DOCUMENT NUMBER	2. DOCUMENT TITLE
MIL-A-12374D	ANEMOMETER ML-433( )/PM AND CASE, ANEMOMETER CY-1397( )/PM

3a. NAME OF SUBMITTING ORGANIZATION	4. TYPE OF ORGANIZATION (Mark one)
	☐ VENDOR
b. ADDRESS (Street, City, State, ZIP Code)	☐ USER
	☐ MANUFACTURER
	☐ OTHER (Specify):

5. PROBLEM AREAS

a. Paragraph Number and Wording:

b. Recommended Wording

c. Reason/Rationale for Recommendation.

6. REMARKS

7a. NAME OF SUBMITTER (Last, First, MI) – Optional	b. WORK TELEPHONE NUMBER (Include Area Code) – Optional
c. MAILING ADDRESS (Street, City, State, ZIP Code) – Optional	8. DATE OF SUBMISSION (YYMMDD)

DD FORM 82 MAR 1426 EDITION OF 1 OCT 76 IS OBSOLETE

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